Problem Statement and Goals
Software Engineering

Team 8, AlgoCatan
Jake Read
Rebecca Di Filippo
Matthew Cheung
Sunny Yao

Table 1: Revision History

Date	Developer(s)	Change
Sept. 22nd	Jake,Matthew	inital problem statement plan created for milestone 1

1 Problem Statement

1.1 Problem

Settlers of Catan is a strategic board game where players act as settlers, competing to build settlements and cities across the map by collecting and trading resources like wheat, wood, brick, etc. It's one of the most popular modern board games, with around 45 million copies sold. It also has a growing competitive scene, complete with Elo ratings in a manner similar to chess. Unlike chess however, there are no existing AI bots capable of playing *Catan* to the level of a human player. This is due to the large state-space of potential moves on a given turn, and the stochastic nature of the game with its dice rolls, as well as only partially observable information (e.g., opponents' hidden cards). This same nature makes it impossible for humans to calculate optimal strategies, as we can't see far enough ahead. A bot capable of playing *Catan* would be useful for both new and experienced players, aiding in learning the complexities of the game. The primary intended use of this project is as a decision-support tool: a human player receives move recommendations and strategic feedback through an application interface. Autonomous full-game play is a secondary use case, intended for self-play training, benchmarking, and optional practice matches against the AI.

1.2 Inputs and Outputs

The system's primary operational inputs are the game state information of an in-progress *Catan* match, including board layout, turn context, and relevant player data needed to explain moves. The system's training inputs are simulated game trajectories produced by a *Catan* rules engine, including self-play episodes and matches against baseline bots, which serve as the main training material for the reinforcement model. The primary outputs are ranked move recommendations and post-game advice, delivered to the user's device to support in-game and post-game decision making. In the secondary autonomous-play mode, the output is the AI-selected action for each turn while acting as a competing player.

1.3 Stakeholders

The primary stakeholders for this project are:

- Players of *Settlers of Catan*: The end-users who will primarily use the AI as an in-game and post-game advisory tool to improve their skills, and secondarily as an autonomous opponent for practice matches.
- Dr. Istvan David: The project supervisor who provides guidance, expertise, and oversight.
- The Project Team: The developers responsible for designing, implementing, and testing the software.
- The Department of Computing and Software (CAS) at McMaster University: The organization hosting the project, which benefits from the academic and technical achievements of its students and faculty.

1.4 Environment

The project requires a hardware and software environment to support its modules. On the software side, the system will need a game simulator for generating training data and evaluating autonomous play, a reinforcement learning framework for the AI and a user-facing application for advisory interaction on the player's device. On the hardware side, the project requires GPUs to train the model, a device with a CPU to run the AI, devices for the players to use the application.

2 Goals

The following are the major goals for the project:

- Create an AI advisory system capable of recommending strong legal *Catan* moves to a human player.

- Support autonomous play as a secondary mode for self-play training, benchmarking, and optional practice matches.
- Accurately record current *static* game state from a physical board.
- Provide valid and reasonable move suggestions to the user.
- Visually presents move suggestions to the user.
- Positive win rate against existing/baseline bot.d

3 Stretch Goals

In addition to the project's primary goals the following are stretch goals we will aim to complete if possible:

- Bot is capable of beating a 1200+ rated *Catan* player (based on existing Elo system).
- Generate human-readable explanations of suggested moves.
- Accurately record *real-time* game state from a physical board.

4 Extras

The extras for this project include:

- User instruction video
- Performance report

Appendix — Reflection

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

Jake Read:

1. Most aspects of this deliverable went quite smoothly. We were able to fairly split the work, in such a way that we all contributed important sections, but were able to work in our own time, reducing time spent in meetings. I believe we were able to balance things in such a way that despite working on different sections, each of us still has a general understanding of the content in each part. What I'm happiest about is that we all seem to be working well together. There hasn't been any conflict between the four of us, and we were upfront about our schedules and availability and have stuck to the expectations we set. Whenever I ran into issues or had questions, it was easy to get a hold of someone in our Discord to help. The Discord server I set up has been perfect for organization, we have a general chat and various channels for sources, notes, resources, etc. Whenever we had any questions none of us could answer, we reached out to our TA or our supervising professor, who were happy to help.
2. We had a couple pain points during the project selection. We knew we wanted to work with AI/ML in the project, but not what project we wanted to do, so we began by running polls in our Discord server on the various potential projects. After voting and a subsequent meeting, we narrowed our options down to two. I was mostly interested in the *Catan* project, while the rest of the team was less certain which they preferred. We decided to schedule a meeting with the supervising profs of both projects, to get a more in-depth idea of what each one involved. This worked, and we ended up going with the *Catan* project. During this whole process, one teammate missed both meetings with little explanation, and was very slow to answer messages. After a discussion with the group, we agreed that we were concerned by the lack of communication, and decided to gracefully part ways with our fifth member.
3. Decisions surrounding scope were quite complex, as none of us had extensive prior experience with reinforcement learning. This made it hard to judge how long certain aspects of the project would take, so we

turned to our supervisor, Professor Istvan David. He was able to give us rough estimates regarding the scope/viability of various goals, which was a great help. The nature of our project made it quite simple to separate goals however, as the design process is rather modular (build simulation, train model, return data, etc.). The existing project description provided in the potential projects document also helped in this regard, as certain milestones were already marked as optional, making them clear contenders for stretch goals.

Rebecca Di Filippo:

1. One thing that went well was how we divided tasks. Each of us focused on our assigned sections individually, and then we reconvened before the deadline to review everything together and compare it against the rubric. This made the process smoother and gave us time to revise before submission. Communication also worked well: using Discord has been especially helpful. We set up polls, channels for resources, and held meetings there, which kept everyone organized and made decision-making easier.

2. The biggest challenge early on was finalizing both a project idea and a stable group. We had several meetings just to settle on which project to choose, and some of the people we initially planned to work with ended up leaving. I had to post multiple times on Avenue and Discord to fill our group, which was frustrating. We also decided to remove one member before the team deadline because they weren't showing up to meetings or contributing. Ultimately, once we solidified the group, things improved. For the project choice, we resolved uncertainty by holding discussions and using Discord polls to weigh different options. Meeting with supervising professors also helped us make an informed decision.

3. At first, we considered including more ambitious features, such as having the agent take in real-time data directly from a physical board using smart glasses or a video stream. After reviewing the overall project scope, we realized these features would be too complex for the time we have. Instead, we focused the core project on developing the reinforcement learning model itself, with simpler inputs like static pictures or descriptions of the game state. More advanced features, such as real-time board capture, multiple camera angles, and post-game explanations generated by an LLM, were moved to stretch goals. This way, the project remains challenging and meaningful, but realistic within the Capstone timeline.

Matthew Cheung:

1. This deliverable was quite straight forward since most of the content I needed to write was already outlined in the potential projects project

description. My writing section was for the problem statement which ended up being paraphrasing and extrapolating on the project description provided. Whatever content I was unsure of was easily cleared up in a call with Istvan David, the prof who wrote the initial project description. However, the best part about this deliverable was the assurance that the team is set on a good trajectory. After meeting with the team and the prof, I'm certain that this will be a great project to work on for capstone. The team is motivated and works well together, and the project has enough depth and complexity to justify working on it for 8 months.

2. A significant initial pain point was the uncertainty surrounding the project's scope and its potential value. We were concerned whether the project was both academically challenging enough for a capstone and if it had sufficient real-world applications to be a valuable addition to our portfolios. Personally, a major worry was that the focus on a board game, combined with the niche nature of reinforcement learning, might not demonstrate skills that are highly sought after in the job market. We resolved these issues through a meeting with our prof Dr. Istvan David. He provided crucial clarity on the depth and breadth of the project. The prof reframed the problem, telling us that it is a serious engineering task that touches upon a wide variety of advanced topics in addition to AI. This discussion addressed our concerns by highlighting that the project, while fun, provides a robust platform for developing and showcasing a diverse set of highly marketable skills.

3. We figured out the scope of our goals with help from our professor and by figuring out what we really wanted to get out of the project. We had a broad idea at first, but after talking with our prof, he gave us a much more detailed, technical breakdown of the problem. This helped us get a way better handle on what was actually realistic to do. We also took a step back and thought about what we wanted to learn from this capstone. Pinpointing the specific skills we were hoping to gain, like working with reinforcement learning and computer vision, let us narrow down our focus. We were able to separate the must-do tasks from the nice-to-have features, making sure our main project was solid and doable, while still leaving room for some cooler stretch goals if we have the time.

Sunny Yao:

1. We were able to clearly define our project objectives and communicate them in a structured way, which helped us align as a team on the overall problem statement. Breaking down tasks early on also made it easier to delegate tasks.

2. A significant initial pain point was the uncertainty surrounding the project's scope and its potential value. We were concerned whether the

project was both academically challenging enough for a capstone and if it had sufficient real-world applications to be a valuable addition to our portfolios. Personally, a major worry was that the focus on a board game, combined with the niche nature of reinforcement learning, might not demonstrate skills that are highly sought after in the job market. We resolved these issues through a meeting with our prof Dr. Istvan David. He provided crucial clarity on the depth and breadth of the project. The prof reframed the problem, telling us that it is a serious engineering task that touches upon a wide variety of advanced topics in addition to AI. This discussion addressed our concerns by highlighting that the project, while fun, provides a robust platform for developing and showcasing a diverse set of highly marketable skills.

3. We figured out the scope of our goals with help from our professor and by figuring out what we really wanted to get out of the project. We had a broad idea at first, but after talking with our prof, he gave us a much more detailed, technical breakdown of the problem. This helped us get a way better handle on what was actually realistic to do. We also took a step back and thought about what we wanted to learn from this capstone. Pinpointing the specific skills we were hoping to gain, like working with reinforcement learning and computer vision, let us narrow down our focus. We were able to separate the must-do tasks from the nice-to-have features, making sure our main project was solid and doable, while still leaving room for some cooler stretch goals if we have the time.